

THE SINK STRENGTH OF A DISLOCATION LOOP IN THE EFFECTIVE MEDIUM APPROXIMATION *

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A method was developed for calculating the sink strength and bias of dislocation loops within the effective medium approximation of Brailsford, Bullough and Hayns. The boundary condition on the toroidal sink surface was replaced by a rate-limiting boundary condition on a sphere enclosing the loop. The effect of the drift field outside the rate-limiting sphere was taken into account explicitly by the use of an effective potential. This scheme was applied to calculate the sink strength and bias of dislocation loops in zirconium. The implications of the results are discussed.

1. Introduction

Dislocation loops constitute one of the main features of the microstructure of irradiated crystalline materials. The nucleation, growth or shrinkage of dislocation loops during irradiation directly reflects the generation, migration and subsequent annihilation at various sinks of the intrinsic point defects created by the irradiation. An understanding of the evolution of the dislocation loops is thus essential to the understanding of the behaviour of the irradiation-generated defects and of the mechanisms that give rise to macroscopic effects such as void swelling, irradiation creep and growth.

Because of the important role of dislocation loop plays in irradiation-induced deformation processes, its function as a point-defect sink has received a great deal of attention. It is generally accepted that dislocation loops are biased sinks, which attract interstitials more than vacancies. In most circumstances, in the absence of sinks of even higher bias, a consequence of this bias leads to a net flow of interstitials to dislocation loops during irradiation. This results in a corresponding supersaturation of vacancies in solution in

the lattice, causing a net flow of vacancies to the less biased sinks, such as grain boundaries and voids. The resulting vacancy–interstitial segregation causes plastic deformation in general. In fact, this is generally accepted to be the basic mechanism responsible for void-swelling due to irradiation [1,2], and has also been proposed as the mechanism for irradiation creep and growth [3,4]. The net flow of interstitials to the dislocation loops will result in the growth of interstitial loops and the shrinkage of vacancy loops. This conclusion has been supported by observation in the case of stainless steel, but is contrary to what has been observed in neutron-irradiation zirconium [5]. Bullough, Hayns and Woo (BHW) [6a] made a first attempt to resolve this apparent contradiction between theory and experiment. These authors calculated the interstitial bias of a dislocation loop as a function of loop radius, r_L , and found that the bias increases with increasing loop size, smaller loops being relatively neutral sinks. Thus, provided that voids are absent, a net flow of vacancies will go to the smaller loops. This competing process provides a sufficient condition for the simultaneous growth of both kinds of loops. If, of course, a vacancy loop reaches a size equal to the neighbouring loops, it will stop growing.

The toroidal geometry of a dislocation loop is

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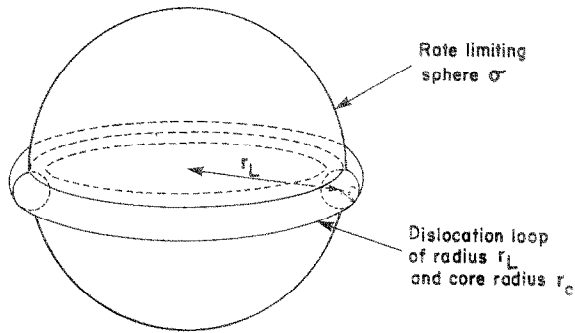


Fig. 1. The circular dislocation loop and the rate-limiting sphere, σ , showing toroidal geometry.

shown in fig. 1. A full drift-diffusional calculation, maintaining this toroidal geometry, is complex. An approximation, adopted by BHW, was to calculate the sink strength of the dislocation loop by an embedding procedure in which the boundary condition at the toroid is replaced by a rate-limiting boundary condition at the surface of an encapsulating sphere, also shown in fig. 1. This is an approach first developed by Brailsford and Bullough [6b]. The transfer velocity, $\bar{K}_L$, across the spherical surface is an adjustable parameter which subsumes both the toroidal geometry and the drift field. This scheme was the basis of the analytical calculation of sink strength as a function of r_L .

Although the essential features of the results of the calculation are probably correct [7], we are aware that the assumption of a constant transfer velocity (independent of r_L), to take into account both the drift field effects and the toroidal geometry of the loop, can cause some inaccuracy in the value of the sink strength so obtained. More accurate values of the sink strength of loops are desirable in rate-theory calculations for direct comparison with experimental observations. Numerical calculations, such as those described in refs. [7–10], are able to provide the required accuracy. In general, however, such calculations are cumbersome for incorporation into the rate theory calculation. This is especially the case when one wants to calculate the sink strength of a dislocation loop within the effective medium approximation [11].

In the present paper, we propose an alternate analytical scheme, in which $\bar{K}_L$ is used to account for

the toroidal boundary condition only. The effects of the drift field outside the rate-limiting sphere are taken into account explicitly by the use of an effective spherical potential [12]. In this scheme, $\bar{K}_L$ is determined by requiring the point defect current due to steady-state diffusion (in the absence of a drift field) across the spherical interface to be equal to that through the toroidal surface of the loop. An analytical expression for the latter has been derived by Seeger and Gösele [13]. With a spherical symmetric region of integration, the angular dependence of the boundary value problem can be tackled by minimizing the Lagrangian with a spherically symmetric trial function, as done by Schröder and Dettmann [12]. This procedure is first applied to calculate the point-defect current into the dislocation loop from a reservoir of point defects, maintained at a concentration $\bar{C}$ at a distance R ($\gg r_L$) away. The error of this procedure is then estimated by comparison with the numerical solution [7] of the same problem, in which the toroidal boundary condition on the surface of the loop, and the full angular dependence of the drift field, are taken into account. The procedure is then applied to derive an expression for the sink strength of a dislocation loop within the effective medium approach. This expression is used to evaluate the sink strength of a dislocation loop in zirconium, as a function of r_L , for various temperatures and dislocation densities. The implications of these results are then discussed.

2. The boundary condition at the dislocation core

One of the difficulties in calculating the sink strength of a dislocation loop is the necessity to solve the drift-diffusion problem with the thermal equilibrium boundary condition *

$$C(r) = C_0(r), \quad r \in \tau, \quad (1)$$

applied on a toroidal dislocation core. Here, $C(r)$ is the point defect concentration at a point r , and

* Note that this is a special case ($v = \infty$) of the boundary condition that Brailsford and Bullough [6b] assumed on the torus, which is given by $-n \cdot J = v(C - C_0(r))$, containing an unknown parameter v , to be fitted to experimental results.

$C_0(r)$ is the thermal equilibrium concentration at r . τ represents the toroidal surface, which has a loop radius r_L and a "pipe radius" r_p . To avoid this difficulty, we envisage that the toroidal boundary condition be replaced by a rate-limiting boundary condition on an effective sphere, σ , of radius r_σ encapsulating the loop (fig. 1). Then we can write

$$I_\sigma = -\frac{\bar{K}_L}{\Omega} \int_{\sigma} [C(r) - C_\sigma(r)] da. \quad (2)$$

Here, I_σ is the current through σ , Ω the atomic volume and da an elemental area on σ . $\bar{K}_L$ and C_σ are functions to be determined. $\bar{K}_L$ plays the role of a transfer velocity that regulates the point-defect current across σ , while $C_\sigma(r)$ is an "effective" thermal equilibrium concentration on σ . In this section, we discuss the way in which $\bar{K}_L$ and C_σ may be chosen to ensure that the toroidal boundary condition (1) can be simulated, to a good approximation, by the spherical one in (2).

A suitable choice of $C_\sigma(r)$ can be derived by considering the steady state flow of point-defects into a dislocation loop, in the absence of point-defect production, from a reservoir maintained at a constant point-defect concentration $\bar{C}$ at a distance $R(>>r_L)$ away from the center of the loop. As shown in Appendix A, the point-defect current into the loop is proportional to $\bar{C} - C^e$ where C^e is given by

$$C^e = C_\infty^e \exp[\pm(\gamma_{SF} + F_{el}/b) \Omega/kT]$$

(see Appendix for meaning of the symbols). Physically, this guarantees that when $\bar{C}$ is equal to the thermal equilibrium concentration, there will be no net flow at steady state into an infinite loop in the absence of an externally applied stress. $C_\sigma(r)$ can be determined by requiring the corresponding current into the effective sphere to satisfy this condition. Thus, the steady-state drift-diffusion boundary value problem (in the effective sphere case) reads

$$\nabla^2(DC) + \beta \nabla E \nabla(DC) + \beta DC \nabla^2 E = 0, \quad (3)$$

subject to the boundary conditions

$$C = \bar{C} \quad \text{at } |r| = R, \quad (4)$$

and eq. (2) on the surface of σ . Here, D is the diffusion coefficient, E is any drift-field and β^{-1} is the product of the Boltzmann constant and the absolute

temperature. By making the transformation

$$\phi(r) = \frac{C \exp(\beta E) - C^e}{\bar{C} - C^e}, \quad (5)$$

the boundary value problem becomes

$$\nabla^2 \phi - \nabla E \nabla \phi = 0, \quad (6)$$

$$\phi = 1 \quad \text{at } |r| = R \quad (7)$$

$$I_\sigma = -\frac{\bar{K}_L}{\Omega} \int_{\sigma} \{[(\bar{C} - C^e) \phi + C^e] \times \exp(-\beta E) - C_\sigma\} da. \quad (8)$$

Using the definition of J , the left-hand side of eq. (8) can be written as

$$I_\sigma = -(\bar{C} - C^e) \int_{\sigma} \exp(-\beta E) \nabla \phi \cdot \mathbf{r} da. \quad (9)$$

This is proportional to $\bar{C} - C^e$ if $\nabla \phi$ is independent of $\bar{C}$ or C^e . A sufficient condition for this to hold can be deduced from eq. (8): it is

$$C_\sigma = C^e \exp(-\beta E_\sigma). \quad (10)$$

Then eq. (8) becomes

$$\int_{\sigma} \exp(-\beta E) \nabla \phi \cdot \mathbf{r} da = \frac{\bar{K}_L}{\Omega} \int_{\sigma} \phi \exp(-\beta E) da. \quad (11)$$

Together with eqs. (6) and (7), this gives a solution ϕ which is independent of $\bar{C}$ or C^e . That eq. (10) is also a necessary condition for the current to be proportional to $\bar{C} - C^e$ is difficult to show for the general case. However, in the case when the drift field E is spherically symmetric, this can be done easily. In this case, the boundary value problem (2)–(4) can be solved and the current through σ is given by

$$I_\sigma = \frac{D 4\pi}{\Omega} \frac{\bar{C} - C_\sigma \exp(\beta E_\sigma)}{W(r_\sigma) - W(R) + \exp(\beta E_\sigma)(D/r_\sigma^2 \bar{K}_L)}, \quad (12)$$

where

$$W(r) = \int_r^\infty \frac{\exp[\beta E(t)]}{t^2} dt, \quad (13)$$

and E_σ is the drift field on σ . Eq. (12) clearly shows that for I_σ to be proportional to $\bar{C} - C^e$, eq. (10) has to be satisfied.

Having established C_σ by eq. (10), we may now attend to the determination of $\bar{K}_L$. $\bar{K}_L$ may be determined by requiring that, in the absence of the drift field and point-defect production, the point-defect current through σ , from a far away reservoir, is the same as that entering the loop. The current I_σ^∞ through σ can be calculated from eqs. (10) and (12), by allowing E_σ to go to zero and R to ∞ . From this, we obtain

$$I_\sigma^\infty = D(\bar{C} - C^e) \frac{4\pi r_\sigma}{\Omega[1 + (D/r_\sigma \bar{K}_L)]} \quad (14)$$

$\bar{K}_L$ can then be determined by equating I_σ^∞ to the current calculated analytically by Seeger and Göselle [13]. From this procedure, $\bar{K}_L$ can be expressed by

$$\bar{K}_L = \frac{2D}{\pi r_\sigma^2} (r_L^2 - r_P^2)^{1/2} \times \left(\frac{Q_{-1/2}(r_L/r_P)}{P_{-1/2}(r_L/r_P)} + 2 \sum_{n=1}^{\infty} \frac{Q_{n-1/2}(r_L/r_P)}{P_{n-1/2}(r_L/r_P)} \right), \quad (15)$$

where $P_{n-1/2}$ and $Q_{n-1/2}$ are, respectively, Legendre functions of the first and second kind of order $n - \frac{1}{2}$ [14]. Eq. (15) defines $\bar{K}_L$ to be used in subsequent calculations. We note that except for the diffusion coefficient D , $\bar{K}_L$ contains only the geometric variables, r_σ , r_L and r_P . This shows that an essential function of $\bar{K}_L$ is to act as a conversion factor for the change of the geometry of the boundary condition from toroidal to spherical. We also note that $\bar{K}_L$ obtained this way only guarantees that the diffusional (field free) current from a reservoir at infinity is calculated correctly. There is no guarantee that it is applicable to cases where the reservoir is at a finite distance R away, nor to cases where a drift-field exists. To test the applicability of $\bar{K}_L$ in eq. (15) to a finite R in the absence of a drift field, we calculate the current I_σ at $R = 80b$ for several values of r_L by solving eqs. (2)–(4) and compare the results obtained by a numerical solution of the corresponding boundary value problem for a toroidal sink using a finite-difference method [7]. The two results are compared in table 1 for different values of r_L . It can be seen that eq. (15) is indeed very accurate.

The case where a drift field exists is considered in the next section.

Table 1

Comparison of the point-defect diffusion current (i.e. no drift) into a toroid of "pipe radius" $r_P = 2b$ from a reservoir $80b$ away maintained at a concentration of $2C^e$ calculated from eq. (4) with that calculated numerically [7]

$r_L(b)$	Current per unit loop length (in units of DC^e/Ω)	
	Numerical	From eq. (4)
5	2.473	2.467
10	2.009	2.005
15	1.863	1.862
20	1.805	1.805

3. Treatment of the drift field

We now consider the calculation of the point-defect migration into the dislocation loop, taking into account the full effects of the drift field, due to the point defect–dislocation interaction. In the absence of the drift flow, and point-defect production, we have shown in the last section that we can replace, with negligible error, the thermal equilibrium boundary condition on the toroidal surface with a rate-limiting boundary condition on a spherical surface σ . In this section, we shall show that the same approximation can be used in the presence of the drift field.

The presence of the drift field complicates the steady state drift-diffusion equation, which now reads [12]

$$\nabla \cdot (D' \nabla \Gamma) = 0 \quad (16)$$

with $D' = D \exp(-\beta E^s)$, in a region where there is no sink, $E^s(\mathbf{r})$ is the point defect–dislocation interaction energy at the saddle point. Here, D is the ideal (i.e. stress-free) diffusion constant, $\Gamma(\mathbf{r})$ is the renormalized point-defect concentration defined by [12]

$$\Gamma(\mathbf{r}) \equiv C(\mathbf{r}) \exp[\beta E^g(\mathbf{r})], \quad (17)$$

where $E^g(\mathbf{r})$ is the point defect–dislocation interaction energy in the stable configuration. The complicated angular dependence of $E^s(\mathbf{r})$ and $E^g(\mathbf{r})$ for a dislocation loop makes the exact solution of eq. (16) extremely difficult, except by numerical means, and then only when the boundary conditions are rela-

tively simple. To reduce this equation to an analytically tractable form, we need a simplifying approximation. First of all following Schröder and Dettmann [12], we define

$$\psi(\mathbf{r}) = -D\Gamma(\mathbf{r}), \quad (18)$$

whence eq. (16) becomes

$$\nabla \cdot [\exp(-\beta E^s) \nabla \psi] = 0, \quad (19)$$

which can be obtained by minimizing the Lagrangian

$$L\{\psi(\mathbf{r})\} = \int \exp(-\beta E^s) (\nabla \psi)^2 d\mathbf{r} \quad (20)$$

with respect to the functional $\psi(\mathbf{r})$ subject to the approximate boundary conditions.

Schröder and Dettmann [12] found that they were able to replace $\psi(\mathbf{r})$ with a spherically symmetric trial solution, $\bar{\psi}(\mathbf{r})$, with negligible error ($\sim 3\%$) in the point defect flux for a spherical sink, even though $E^s(\mathbf{r})$ is non-spherically symmetric.

We can adopt the same approximation here and replace the differential equation (19) with

$$\frac{d}{dr} \left(r^2 g(r) \frac{d\bar{\psi}}{dr} \right) = 0. \quad (21)$$

Here, $g(r)$ is an effective spherical potential given by

$$g(r) = \frac{1}{4\pi} \int \exp[-\beta E^s(\mathbf{r})] d\omega \quad (22)$$

with the integration carried over all solid angles ω .

Replacing the toroidal boundary condition (1) and the non-spherically symmetric drift-diffusion equation (19) by the spherically symmetric boundary condition (2) and drift-diffusion equation (21), the calculation becomes much simplified.

Before proceeding further, let us estimate the magnitude of errors involved in this spherical symmetrization (SS) procedure. We do this by adopting the SS procedure to calculate the point-defect current into the dislocation loop from a reservoir maintained at a renormalized point-defect concentration $\Gamma(\mathbf{r}) = \bar{\Gamma}$ at a distance $R(\gg r_L)$ away from the center of the loop, and comparing the results with that obtained from a finite-difference numerical solution (7) of the same problem without spherical symmetrization [i.e. eqs. (1) and (19) are not replaced by eqs. (2) and (21)].

To avoid unnecessary complication, we made the

simplifying assumption that the defect strength in the saddle point configuration, and thus the interaction potential, is the same as the stable configuration. Then, we can put $E^s = E = E^g$. We emphasize that this assumption is made only to reduce the algebraic complexity, not for any other specific reason.

After the SS procedure, the boundary value problem reads

$$\frac{d}{dr} \left(r^2 g(r) \frac{d\bar{\psi}}{dr} \right) = 0, \quad (23)$$

$$\bar{\psi}(r) = -D\bar{C} \quad \text{on the spherical reservoir } |r| = R, \quad (24)$$

$$I = -\frac{\bar{K}_L}{\Omega} \int_{\sigma} [C(\mathbf{r}) - C_0(\mathbf{r})] da, \quad (25)$$

where C and $\bar{\psi}$ are related by

$$\bar{\psi}(r) = -DC(r) \exp[\beta E(r)], \quad (26)$$

and

$$C_0 = C^e \exp[-\beta E(r)]. \quad (27)$$

The solution can be obtained readily and the current is given by

$$I = \frac{4\pi}{\Omega} D \frac{\bar{C} - C^e}{X(r_0) - X(R) + Dg(r_0)/(r_0^2 \bar{K}_L)}, \quad (28)$$

where

$$X(r) = \int_r^{\infty} \frac{dt}{t^2 g(t)}. \quad (29)$$

For a circular edge dislocation loop in an anisotropic medium, its interaction energy with a point-defect, simulated by a spherical inclusion of radius $r_0(1 + \epsilon)$ in a spherical cavity of radius r_0 , has been calculated by Bastecka and Kroupa [15]. When the point defect is located at a point with cylindrical co-ordinates (r, ϕ, z) , as depicted in fig. 2, the interaction energy, $E(r, z)$, is independent of ϕ and is given by

$$E(r, z) = -\frac{4}{3} \frac{1 + \nu}{1 - \nu} \mu r_0^3 b \epsilon [(r + r_L)^2 + z^2]^{-1/2} \\ \times \left(\frac{r_L^2 - r^2 - z^2}{(r_L - r)^2 + z^2} \right) E_e(\chi) + K_e(\chi), \quad (30)$$

where ν is Poisson's ratio, μ is the shear modulus, b is

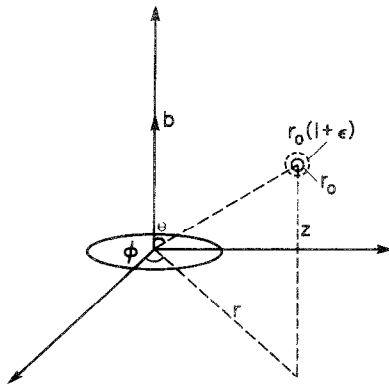


Fig. 2. The circular pure edge-dislocation loop and the disposition of the co-ordinate system used. The location of the point defect is also indicated.

the Burgers vector, and $E_e(\chi)$, $K_e(\chi)$ are complete elliptic integrals [14] of the first and second kinds, respectively. The argument χ is defined by

$$\chi = \frac{4rr_L}{(r + r_L)^2 + z^2}.$$

The point-defect current per unit dislocation line length calculated in this way is plotted as a function

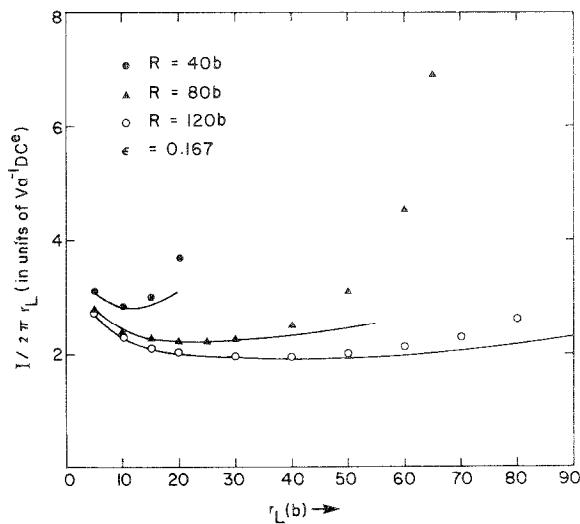


Fig. 3. Comparison of the point-defect currents per unit loop length by the present calculation (solid lines) with those obtained numerically by a finite-difference method [7] (discrete symbols) for different loop radii r_L and different distance R from the surrounding spherical reservoir. Here $\epsilon = 0.167$.

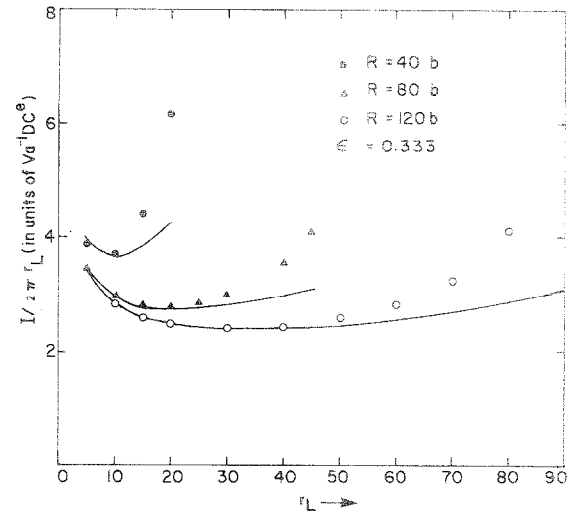


Fig. 4. Same as fig. 3, but with $\epsilon = 0.333$.

of r_L in figs. 3 and 4 for $\epsilon = 0.167$ (volume strain ≈ 0.6) and $\epsilon = 0.33$ (volume strain ≈ 1.35) respectively. This is compared with the numerical results of a finite difference calculation of the same problem without the SS procedure by Woo et al. [7]. The corresponding solutions $\bar{\psi}(r)$ and $\psi(r)$ for $R = 80b$ and $r_L = 20b$ are compared in fig. 5. The materials parameters are those appropriated to zirconium and are given in table 2.

The three cases under consideration cover a loop number density of $4 \times 10^{21}/\text{m}^3$ ($R = 120b$) to $1.1 \times 10^{23}/\text{m}^3$ ($R = 40b$). It can be seen that the analytical approximation is accurate for the smaller loops, in all cases. However, as the loop radii increase, the analytical approximation loses accuracy. This can be rationalized as follows: in the analytical approximation, the effect of the drift field inside the rate-limiting sphere is not accounted for at all, except in the immediate neighbourhood of the dislocation core, where it is taken into account by making the pipe radius equal

Table 2
Materials data for zirconium used in the calculation described in section 5

ν	μ (dynes/cm ²)	b (nm)	Ω (m ³)
0.33	33×10^{11}	0.323115	23.27×10^{-23}

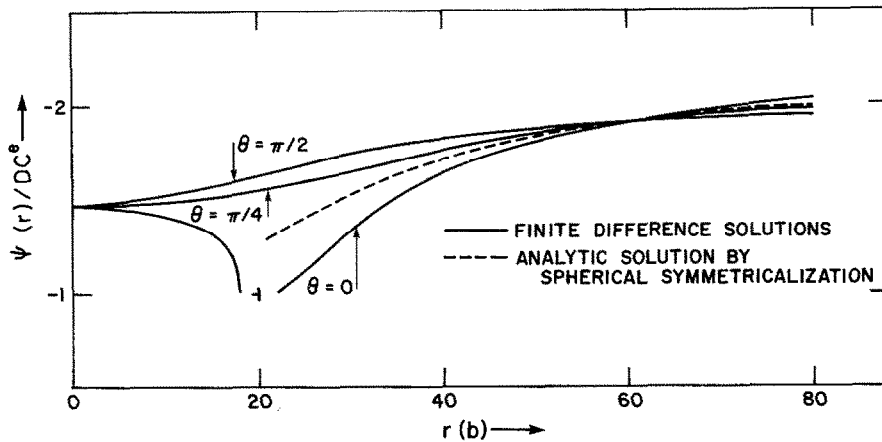


Fig. 5. Comparison of the solution $\psi(r)$ calculated by a finite-difference method [7] with $\bar{\psi}(r)$ calculated by spherical symmetrization. θ is the angle as depicted in fig. 2.

to the effective capture radius (see below). As the rate-limiting sphere increases in size, the region in which the drift flow is neglected becomes large compared with the region in which the drift flow is included. In addition, the accuracy of the effective potential approximation decreases for larger loops because ψ becomes more angular dependent under such circumstances.

In order to minimize this error, we have chosen r_σ to be the minimum radius for the rate-limiting sphere which can encapsulate the loop completely, i.e.

$$r_\sigma = r_L + r_p. \quad (31)$$

Here, r_p is the pipe radius of the effective toroidal surface on which the boundary condition (1), for the finite-difference calculation, is defined. This surface encloses a region in which a newly arrived point defect will find it difficult to leave by thermal motion and thus may be considered to be absorbed. The choice of r_p is a difficult task. We follow Schröder and Dettmann [12] and define r_p as the pipe radius of the surface on which $E^g(\infty) - E^g(r) = kT$. The precise definition of r_p here is not very important because we are comparing the results of two calculations that use the same r_p .

Although the analytical approximation loses accuracy as the loop radius gets large, it should be noted that the scheme for modelling the point-defect migration into a dislocation loop during irradiation, by means of the drift-diffusion from a spherical

reservoir (assumed in the numerical solution [71]) also loses validity when the loop approaches the reservoir. Under such circumstances, the drift-diffusion from a toroidal reservoir surrounding the loop is perhaps a more appropriate model. The pipe radius, R_p , of such a reservoir may be defined by requiring that the volume of material contained in the encapsulating toroid be equal to that enclosed by the spherical reservoir, so that the same loop line density is maintained in both cases. The R_p that satisfies this condition is given by

$$R_p = (2R^3/3\pi r_L)^{1/2}. \quad (32)$$

For large loops, the curvature of the dislocation line is negligible. The point-defect current into a unit length of the loop, Z , may be approximated by that from an equivalent straight dislocation surrounded by a cylindrical reservoir. This has been calculated by Heald [16]. The results are plotted in fig. 6, for the case where $R = 80b$ and $\epsilon = 0.167$. The equivalent currents computed for the case with the spherical reservoir are also plotted for comparison. It can be seen that the two results start to deviate significantly when $r_L \geq 50b$. We note that, at this point, $R_p \approx r_L$ and the geometry of the two reservoirs becomes radically different. This observation enables one to state a criterion for the range of applicability of the results derived from the numerical computation to irradiation damage studies. A reasonable limiting criterion is $r_L < 0.8R_p$, which is equivalent to $r_L <$

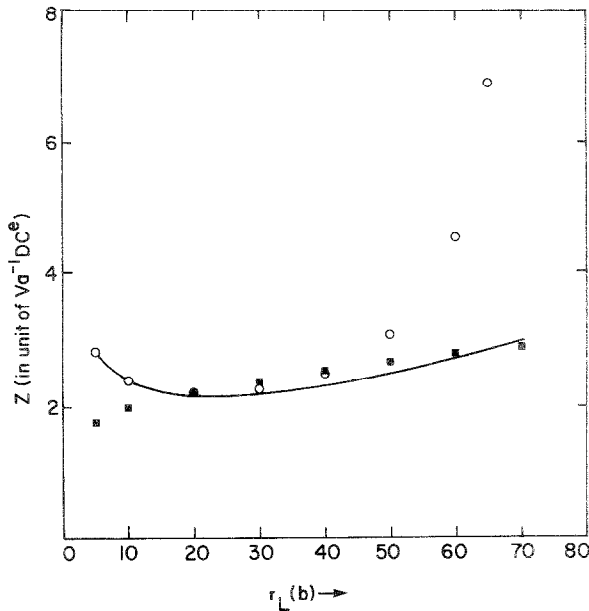


Fig. 6. Point-defect current per unit loop lengths versus loop radius r_L for a loop number density of $1.37 \times 10^{22}/\text{m}^3$ (corresponding to $R = 80b$) and $\epsilon = 0.167$. Comparison among the numerical solution with spherical reservoir ($\circ$), the straight dislocation model ($\blacksquare$) and the present calculations (solid line).

$0.5R$, according to eq. (32). Referring to figs. 3 and 4, it can be seen that the analytical approximation is also quite accurate in this range. In fig. 6, we also compare the analytic solution with the straight dislocation model. It is seen that the two models agree rather well, except for very small loops, where the straight dislocation model is obviously invalid. From this comparison, one may conclude that the analytic approximation is applicable over the full range of r_L . However, for $r_L > R/2$, such an agreement may be due to a fortuitous cancellation of errors, which may not happen under other circumstances. Therefore, it would be safer to simply apply the straight dislocation model in this range.

4. The effective-medium approach

To avoid additional complications at this stage, we consider the situation where dislocation loops of a unique size constitute the only sinks. As in ref. [11], the sink strength of the equivalent effective medium

is then calculated from the point-defect flux into a discrete sink surrounded by a concentric annular spherical zone (of radius R), within which there are no other sinks. Outside this zone the effective medium begins and extends to infinity.

In the case being considered here, continuous point-defect generation creates a supersaturation in the matrix that drives the point-defects into the loop, just as the surrounding reservoir discussed in the last two sections does (except that the "reservoir" here has no definite boundary). We have shown in the last two sections that such migrations can be calculated sufficiently accurately by the SS procedure. Of course, one must realize that, with point-defect generation, the current through σ is not the same as that going into the loop (see Appendix B). Then, if this fact can be accounted for, we may adopt the SS procedure to calculate the sink strength of the loop within the effective medium approximation.

Following Brailsford et al. [11], Maxwell's approach is used to fix the radius of the sink free region

$$R = \left(\frac{4}{3}\pi N_L\right)^{-1/3}, \quad (33)$$

where N_L is the loop number density. Using the SS procedure, the BVP for point-defect migration in the volume bounded by σ and the effective medium, is given by

$$\left\{ \begin{array}{l} \frac{1}{r^2} \frac{d}{dr} \left[r^2 g(r) \frac{d}{dr} \bar{\psi}(r) \right] - K = 0, \quad r_\sigma < r \leq R, \quad (34) \\ \frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{d}{dr} \bar{\psi}(r) \right] - K' - k^2 \bar{\psi} = 0, \quad r > R, \quad (35) \end{array} \right.$$

$$\bar{\psi} \text{ and } \bar{\psi}' \text{ are continuous at } R, \quad (36)$$

$$I_\sigma = -\frac{\bar{K}_L}{\Omega} \int_\sigma (C - C_0) d\sigma. \quad (37)$$

Here, K' is the sum of K and K^e , which are, respectively, the point-defect production rate (in dpa/s), and the point-defect thermal emission rate. k^2 is the sink strength of the equivalent lossy medium. I_σ is the current through the rate-limiting sphere, σ , defined in section 2. In formulating the above equations, we have assumed that R is large enough that the interaction energies are negligible for $r > R$.

Solving this set of equations and requiring equivalence between the effective medium and the discrete sink, we obtain (see Appendix B)

$$K_e = k^2 DC^e, \quad (38)$$

and

$$k^2 = (4\pi N_L) \times \left(\frac{4\pi N_L}{3} [F(r_o) - F(R)] + X(r_o) - X(R) + \frac{4\pi N_L \gamma}{3r_o h(r_o)} (R^3 - r_o^3) \right)^{-1}, \quad (39)$$

where $F(r)$ and $X(r)$ are defined by

$$F(r) = \int_0^r \frac{t}{g(t)} dt, \quad (40)$$

$$X(r) = \int_r^\infty \frac{dt}{t^2 g(t)}, \quad (41)$$

$$\gamma = D/r_o \bar{K}_L, \quad (42)$$

and $h(r)$ is defined similar to $g(r)$ in eq. (22), but with $E^s(r)$ replaced by $E^B(r)$. In the present paper we assume that $E^s(r) = E^B(r)$ and hence $g(r) = h(r)$.

Although eq. (39) is derived for a dislocation loop, it can be shown that it is valid for any sink with a spherical rate-limiting surface of radius r_o . If we assume that $r_o \ll R$ and the point-defect–void interaction energy is negligible, we obtain the usual expression for the sink strength of a void in the absence of other sinks, namely

$$k^2 = 4\pi r_s N_L,$$

in the diffusion control limit when $\gamma \rightarrow 0$.

Eq. (39) gives the sink strength of a dislocation loop in the effective-medium approximation, with the effect of the drift field taken into account explicitly. Comparison with the finite-difference calculation [7] indicates that this expression should be accurate for $r_L < R/2$.

5. Sink strength and bias of dislocation loops in zirconium

Using eq. (39), with the interaction energy given by eq. (30), the sink strengths per unit line length, Z ,

and bias of dislocation loops, $B(\equiv 1 - Z_v/Z_i)$, can be calculated readily by standard numerical procedures. This is done for zirconium, using the materials parameters in table 2. Unfortunately, the two crucial parameters, ϵ_i and ϵ_v , for the relaxation strains of the interstitials and vacancies respectively, are not known. Calculations are therefore done for a range of ϵ_i and ϵ_v , in which the ratio ϵ_i/ϵ_v satisfies $1.5 < |\epsilon_i/\epsilon_v| < 3$ and $0.083 < |\epsilon_v| < 0.167$, corresponding to a vacancy relaxation volume strain between 27 and 60%.

In general, the results show that, for any fixed loop radius r_L and fixed loop number density N_L , Z increases with $|\epsilon|$ for both types of defects, and B increases with the ratio $|\epsilon_i/\epsilon_v|$. Both results are expected because, for a fixed r_L , the sink strength should increase with the strength of the drift field. In addition, we also examine the functional dependence of B on $|\epsilon_v|$ and the $|\epsilon_i/\epsilon_v|$ ratio. Accordingly, in fig. 7, we plot B versus r_L for a loop number density $10^{22}/\text{m}^3$ at 500 K with $|\epsilon_v| = 0.167$ for different $|\epsilon_i/\epsilon_v|$ ratios. To show the effects of changing $|\epsilon_v|$, we also plot on the same figure the corresponding graphs for $|\epsilon_v| = 0.083$ in broken lines. From fig. 7, it can be seen that for a fixed $|\epsilon_i/\epsilon_v|$ ratio, B is a moderately increasing function of $|\epsilon_v|$. In the case considered in fig. 7, doubling $|\epsilon_v|$ from 0.083 to 0.167, only results in an increase of B of about 20–30%. The functional dependence of B on the $|\epsilon_i/\epsilon_v|$ ratio, on the other hand, is much stronger. This is illustrated in fig. 8,

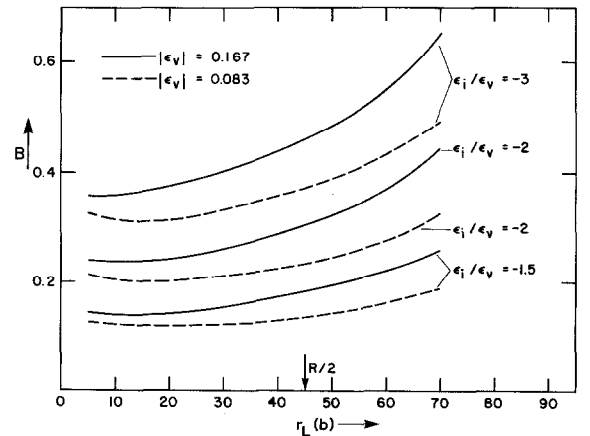


Fig. 7. Comparison of B as a function of r_L for different ratios of ϵ_i/ϵ_v and $|\epsilon_v|$ at $T = 500$ K and $N_L = 10^{22}/\text{m}^3$.

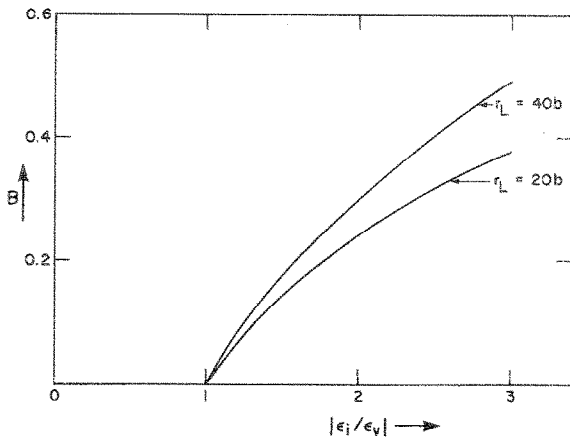


Fig. 8. The bias B as a function of $|\epsilon_i/\epsilon_v|$ for two different loop radii.

where we plot B versus $|\epsilon_i/\epsilon_v|$ for two values of r_L , $20b$ and $40b$.

From an analysis of the experimental void-swelling results of zirconium irradiated in the high voltage electron microscope (HVEM), Faulkner and Woo [17] estimated a dislocation bias (mainly due to loops) of 0.21. Buckley et al. [18] found that a bias of 0.25 is required to explain the irradiation growth of zirconium and Zircaloy under electron irradiation. From fig. 8 then, a value of $|\epsilon_i/\epsilon_v| \approx 2$ is perhaps a good estimate for zirconium. Although at this stage, the effect of temperature and loop number density on the functional dependence of B on $|\epsilon_i/\epsilon_v|$ has not been taken into account, later results will show that, within the range of temperature (500–800 K) and loop number density (10^{20} – $10^{22}/\text{m}^3$) investigated experimentally, these effects are moderate and $|\epsilon_i/\epsilon_v| \approx 2$ is a good estimate for zirconium. Lastly, we note in fig. 7 that despite the rather strong dependence of B on $|\epsilon_i/\epsilon_v|$, the shape of the B versus r_L plots varies only moderately with $|\epsilon_i/\epsilon_v|$.

The foregoing arguments allow us to base the following discussion on the assumption that $|\epsilon_i/\epsilon_v| = 2$ and $\epsilon_v = 0.167$ for zirconium. We believe that deviations of the actual values from the assumptions do not affect the essentials of our conclusions.

In the following, our discussion centres on the functional dependences on r_L of both Z and B , and how temperature and loop number density affect them.

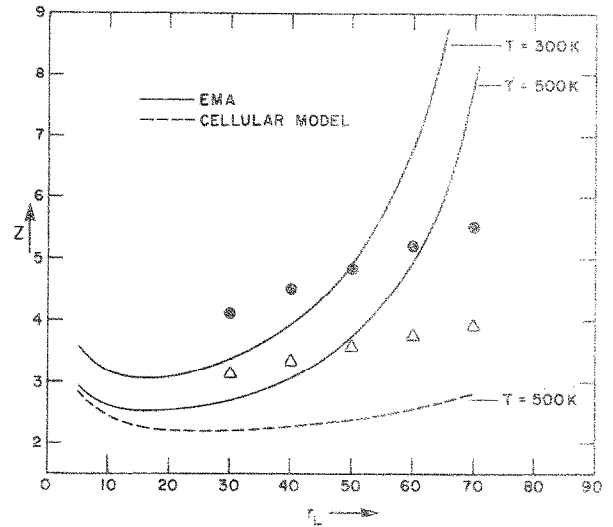


Fig. 9. Vacancy current per unit loop length Z versus loop radius r_L for $N_L = 10^{22}/\text{m}^3$ for two temperatures 300 and 500 K (solid line). Plotted in comparison as discrete points are the Z 's calculated by the corresponding straight dislocation model in eq. (43). For comparison, the corresponding Z for $T = 500$ K calculated by Heald's approach [16] is plotted as a broken line.

The resulting Z for the vacancy is plotted against r_L in fig. 9 as solid lines for two temperatures, 300 and 500 K, and $N_L = 10^{22}/\text{m}^3$. For this loop number density, $R = 89b$. As the loop radius increases from $5b$, the value of Z decreases, passes through a minimum and then increases monotonically. This general behaviour is typical for the range of temperature and loop density under consideration and is similar to the results of Bullough et al.'s finite difference calculation [19]. The position of the minimum is insensitive to temperature but is sensitive to the loop number density. As the loop number density decreases, the position of the minimum moves towards larger r_L . In fact, when the loop number density approaches zero, the minimum moves to infinity. This corroborates the results of Wolfer [20] which show that Z is a monotonic decreasing function of r_L when the reservoir is at ∞ . The behaviour of Z as a decreasing function of r_L at small r_L is characteristic of the point-defect diffusion to a toroid, as can be seen from the finite-difference results in table 1 and also the analytic results of Seeger and Gössele [13]. As r_L becomes larger, Z starts to increase with r_L . This behaviour of Z is the result of two factors. The most important of

these is probably the proximity of the dislocation line to the effective medium. Strictly speaking, surrounding the dislocation loop by a spherical sink-free region of radius R is valid only when the loop radius is much smaller than the radius of the sink-free region. Otherwise, a toroidal sink-free region surrounding the loop is perhaps a more appropriate model. The spherical sink-free region causes the dislocation line to come too close to the effective medium when the loop radius approaches R . This results in a large overestimation of the average distance between neighbouring loops and hence a large overestimation of the sink-sink interaction. The rapid increase of Z as r_L approaches R in fig. 9 is a reflection of this fact. Another factor which increases Z is that, as r_L increases, there is less cancellation between the strain fields of the opposing segments of the loop. Consequently, as r_L increases, the drift field also increases, thus causing an increase in Z .

The foregoing discussion suggests the possibility of an overestimation of the values of Z when r_L approaches the value of R , because of the inappropriate use of a spherical sink-free region. Nevertheless, following the arguments of section 3, this problem would not be too serious if the use of spherical sink-free region is restricted to cases where $r_L < R/2$. For larger values of r_L , we suggest that the effect of the curvature of the dislocation line may be neglected and the use of a straight dislocation model is more appropriate. Using the spherical-symmetrization approach* in the present paper, an expression for the corresponding straight dislocation sink strength can be derived and is given by

$$k^2 = \{4\pi\rho\} \times \left\{ \left(1 + \frac{L^2}{4R_p^2}\right) \ln \left[\frac{\left(R_p^2 + \frac{L^2}{4}\right)}{\left(b^2 + \frac{L^2}{4}\right)} \right] - \left(1 - \frac{b^2}{R_p^2}\right) \right\}^{-1}, \quad (43)$$

* The applicability of this approach to the case of straight dislocations has been checked in the case of point-defect diffusion from a surrounding reservoir, where the exact solution is available [16]. The present approach in general underestimates the flux by about 2%. It can therefore be considered quite accurate.

where

$$L = \frac{4}{3} \frac{1+\nu}{1-\nu} \mu r_0^3 b \epsilon, \quad (44)$$

and R_p is the pipe radius of the toroidal sink-free region, related to R through eq. (32).

Corresponding results calculated using eq. (41) are also plotted in fig. 9 as discrete points. It can be seen that, for $r_L \gg R/2$, the results from the two models deviate quite significantly. This confirms the inappropriateness of the eq. (39) in this regime. In the neighbourhood of $r_L = R/2$, on the other hand, the two models produce results close to each other. Since the geometries of the sink-free regions of both models are quite similar in this case, this shows that the curvature of the loop is indeed negligible when $r_L > R/2$, and that either model is applicable in this regime. This allows an interpolation scheme to connect the two models.

The present method also allows a direct comparison between the Z 's calculated by the effective-medium approach in eq. (39) and Heald's approach where continuous defect production is not accounted for. Corresponding results of the latter for a temperature of 500 K are plotted also in fig. 9 as broken lines. Comparison with the effective medium results shows that Heald's approach [16] always underestimates the value of Z , especially in the regime where the dislocation line is close to the effective medium, i.e., in cases where the loop line density is high and sink-sink interaction is very significant. On the other hand, when the loops are small and the loop dislocation line is far from the effective medium, the error of Heald's approach is small.

We now consider the dependence of the bias B on the loop radius r_L and how temperature and loop number density affects the relationship. B , as a function of r_L , is plotted in fig. 10 as solid lines for a temperature of 500 K and loop number densities of $N_L = 10^{21}/\text{m}^3$ and $10^{22}/\text{m}^3$. The discrete points are calculated from the straight dislocation model. The corresponding results for 700 K are presented in fig. 11.

The general features of all the curves under consideration are similar. As r_L increases, B first decreases, passes through a shallow minimum and increases again for larger values of r_L . The increasing behaviour of B with r_L for large r_L can be understood

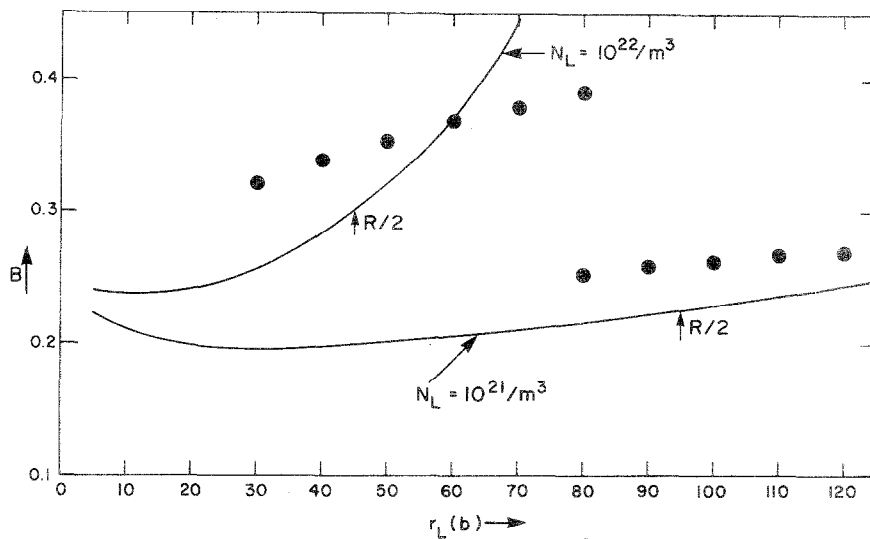


Fig. 10. B as a function of r_L for $T = 500$ K and $N_L = 10^{21}/\text{m}^3$ and $10^{22}/\text{m}^3$. The discrete points are calculated from the straight dislocation model.

as follows. As r_L increases, the dislocation line density also increases. This causes an increase in the sink-sink interaction and thus an increase in Z . Now, the sink-sink interaction increases with the strength of the drift field; this means that the increase in Z for the interstitials is larger than that for the vacancies. The final result is a bias which is an increasing function of r_L .

When r_L is small, however, the above effect is not dominant. Instead, the rapidly changing effective capture radius, r_p , of the dislocation core of the loop plays the essential role. The variation of r_p with r_L can be expected from physical considerations. As the loop gets smaller, there is an increasing cancellation of the dislocation strain field and a consequent weakening of the drift field, which causes r_p to

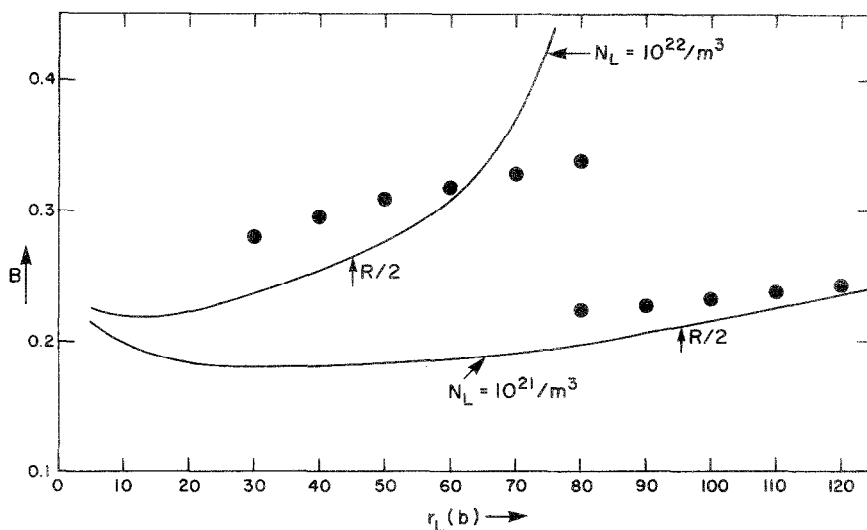


Fig. 11. Same as fig. 10 but for $T = 700$ K.

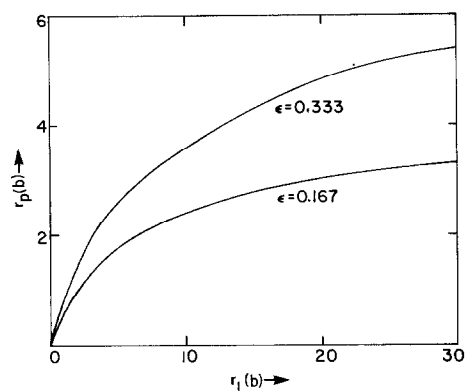


Fig. 12. The effective capture radius r_p versus r_L . Note the rapid decrease of r_p when the size of the loop decreases.

decrease. In fig. 12, we plot r_p versus r_L , calculated by the method as described earlier in section 3, for both the vacancy ($|e| = 0.167$) and interstitial ($|e| = 0.333$). It can be seen that r_p indeed decreases as the loop size reduces, and at small r_L , the decrease is quite significant. The change in r_p , in response to the change in the loop size, induces a change in Z . Comparison of the results of Seeger and Gösele [13] for field-free diffusion and the finite-difference calculation of Woo et al. [7] for drift diffusion, suggests that the effect of r_p on Z decreases as the drift field increases. In fact, Woo et al. [7] showed that, going from a pipe radius of $3b$ to $2b$ causes a decrease of Z which is twice as much in the case of vacancies as for interstitials. The same effect is probably more obvious in the straight dislocation model. Thus, referring to eq. (43), when L is large, changing the value of $b (=r_p$ in this case) does not affect the value of k^2 . However, when L is small, changing the value of b will have a much bigger effect. Thus, as the loops become smaller, the Z for the vacancies decreases faster than that for the interstitials. Consequently, the bias increases as r_L decreases for the smaller loops. This explains the difference between the present results for B and those obtained by Bullough et al. [19] who chose to use a fixed capture radius of $2b$ and thereby obtained B as a monotonic increasing function (also see figure 12 of ref. [7]).

Comparison of figs. 10 and 11 shows that, apart from reducing the bias by 10–20%, raising the temperature from 500 to 700 K does not affect the shape of the

functional dependence of B on r_L very much. The dependence of B on the loop number density, on the other hand, is much more significant. The influence of the loop number density on the bias is mainly effected through the change in the sink–sink interaction. Thus, when the loops are small, each individual loop is essentially isolated; increasing the loop number density only increases B slightly. However, when the loops get larger, increasing the loop number increases the sink–sink interaction significantly. This causes Z , and hence B , to increase, as discussed earlier.

The most important observation from figs. 10 and 11 is perhaps the simple fact that the bias of a dislocation loop varies with the loop size. In the absence of neutral sinks, such as voids and grain boundaries, this bias differential would cause a net interstitial flux to loops with higher bias and a net vacancy flux to those with lower bias. This allows the possibility of vacancy loop growth in zirconium, where void growth seldom occurs. This is an experimental observation [5] which was puzzling until BHW offered the first explanation. Based on the last jump model, which neglects the effects of the loop size on the transfer velocity, a bias was derived which increases monotonically with r_L . With this model, vacancy loop growth can readily be explained. The collapse of the vacancy-rich cores of the displacement cascades presents a situation in which small vacancy loops are continuously being created in the presence of a background microstructure. In the simple case where the microstructure is dominated by a dislocation network, the higher bias of the network guarantees a net flow of vacancies to the small vacancy loops thus causing them to grow. In the absence of a dominating network, as in annealed material, the cascade collapse causes a larger interstitial supersaturation and hence the condition for the interstitial loops to grow. When a background microstructure made up of larger interstitial loops is developed, the higher bias of the larger interstitial loops again causes growth of the smaller vacancy loops.

The main difference between the r_L -dependence of the bias obtained by BHW and that presented here occurs in the regime of small r_L . According to BHW, the value of the bias is $\approx \frac{4}{3}B_\infty$ when r_L approaches zero, where B_∞ is the bias for a straight dislocation. This is rather different from the present results, as

can be seen in figs. 10 and 11. The main reason for this difference may be the breakdown, in the analysis of BHW, of the assumption of a constant transfer velocity $\bar{K}_L$. Nevertheless, the present results do not contradict the essentials of the vacancy-loop growth mechanism proposed in that paper, because of the existence of a similar bias differential between the larger and the smaller loops. Additionally, the present results are consistent with the observed variation of the extent to which vacancy loop growth occurs at different temperatures. Thus, at lower temperatures, it is generally observed that the loop number density is high and the loop radii are small. Typical experimental measurements for zirconium [5] show that most of the loops, say at 500 K, have radii less than $30b$ and a loop density of $10^{22}/\text{m}^3$. From fig. 10 it can be seen that, under such conditions, the bias differential would be small for annealed materials, and vacancy loop growth would not occur to any large extent. This is consistent with experimental observations [5]. However, if the initial microstructure is dominated by the dislocation network, the present theory predicts that vacancy loop growth would be extensive even at the lower temperature. This prediction could be tested experimentally. At a higher temperature, on the other hand, the loop number density is low. Even if a network is absent initially, the interstitial loops can grow to a large size and supply the necessary bias differential to cause vacancy loop growth. In such a case, the vacancy loops will grow to sizes comparable to those of the interstitial loops.

The foregoing discussions on microstructural evolution are qualitative and conceptual only; further rigorous calculations are needed to elaborate the theory and to explore other consequences related to the r_L -dependence of the bias. We also note that we have neglected the influence on the sink strength due to the internal stress from other loops, which would be significant at high sink density case. However, such calculations are necessarily complicated and are outside the scope of the present paper.

6. Summary and conclusions

A method was developed for calculating the sink strength and bias of dislocation loops within the effective-medium approximation of Brailsford, Bul-

lough and Hayns [11]. The thermal equilibrium boundary conditions on the toroidal sink surface were replaced by a rate-limiting boundary condition on a sphere enclosing the loop. The effects of the drift field outside the rate-limiting sphere was taken into account explicitly by the use of an effective potential [12].

This scheme (called spherical symmetrization procedure) was first applied to calculate the point-defect current into the dislocation loop from a reservoir surrounding the loop at a distance R away. The accuracy of the procedure was then evaluated by comparing with the exact results without spherical symmetrization, from a finite-difference numerical calculation [7]. Good agreement was found between the two solutions when the loop radius $\leq R/2$. Reasons for this were discussed.

The scheme was then applied to derive an expression for the sink strength of dislocation loops within the effective-medium approximation. The expression was evaluated using material parameters specific to zirconium. The sink strength per unit line density, Z , and bias, B , were examined as a function of the loop radius for various temperatures. The following conclusions were obtained:

(1) B is a strongly increasing function of the ratio $|e_i/e_v|$ and a weaker increasing function of $|e_v|$. A value of $|e_i/e_v| \approx 2$ was found to be necessary to obtain a bias of ~ 20 – 30% for zirconium, a value estimated from experiments.

(2) For a fixed loop number density and temperature, as the loop radius increases from $5b$ the values of both Z and B first decrease, then pass through a minimum and finally increase monotonically. The position of the minimum is insensitive to temperature, but is sensitive to the loop number density.

(3) For a fixed loop number density and loop radius, both B and Z are decreasing functions of temperature.

(4) For a fixed temperature and loop radius, both B and Z are increasing functions of loop number density. The rate of increase is larger for bigger loops.

(5) The sink strength of a dislocation loop calculated from a model involving a surrounding spherical sink-free region of radius R is a valid approximation only when the loop radius is smaller than $R/2$. A straight dislocation model is adequate for larger loops.

(6) Despite the difference between the present results and those of BHW, the vacancy-loop growth mechanism proposed by BHW, namely, that a bias differential exists between the larger and the smaller loops, is still valid.

Appendix A

In this Appendix, we consider the steady state point-defect current into a dislocation loop, from a reservoir maintained at a constant point-defect concentration $\bar{C}$ at a distance $R(>>r_L)$ away from the center of the loop. We shall show that the current into the loop is proportional to $\bar{C} - C^e$, where C^e is given by

$$C^e = C_\infty^e \exp[\pm(\gamma_{SF} + F_{el}/b) \Omega/kT]$$

with C_∞^e the thermal equilibrium concentration far away from the dislocation, γ_{SF} the stacking fault energy for faulted loops (otherwise $\gamma_{SF} = 0$) and F_{el} the elastic energy (line tension) of the loop. For vacancy emission, the plus-sign is used with vacancy loops, and the minus-sign with interstitial loops (vice versa for interstitial emission).

By making the transformation

$$\psi(r) = -DC(r) \exp\{\beta E(r)\}, \quad (A.1)$$

where $E(r)$ is the drift field, the drift-diffusion boundary value problem can be written as [7]

$$\nabla^2 \psi - \nabla E \cdot \nabla \psi = 0, \quad (A.2)$$

$$\psi = -D\bar{C} \quad \text{at } |r| = R \quad (E \approx 0), \quad (A.3)$$

$$\psi = -DC^e \quad \text{on } \tau. \quad (A.4)$$

If we further transform

$$\phi = \frac{\psi + DC^e}{DC^e - D\bar{C}}, \quad (A.5)$$

the boundary value problem becomes

$$\nabla^2 \phi - \nabla E \cdot \nabla \phi = 0, \quad (A.6)$$

$$\phi = 1 \quad \text{at } |r| = R, \quad (A.7)$$

$$\phi = 0 \quad \text{on } \tau. \quad (A.8)$$

The solution $\phi(r)$ of (A.5)–(A.7) is independent of C^e and $\bar{C}$.

From eq. (A.5), the solution of (A.2)–(A.4) can be written as

$$\psi(r) = -D(\bar{C} - C^e) \phi(r) + DC^e. \quad (A.9)$$

The point-defect current I into the loop is given by

$$I = - \int_{\Sigma} \exp(-\beta E) \nabla \psi \cdot \mathbf{n} \, da, \quad (A.10)$$

where the integration is carried out over any closed surface Σ with normal $\mathbf{n}$ encapsulating the loop and no other sources or sinks. It follows from (A.9) and (A.10) that

$$I \propto \bar{C} - C^e. \quad (A.11)$$

We note that this general result is consistent with the particular result for a straight edge dislocation obtained by Heald (see equation (18) of ref. [16]). Furthermore, it is also consistent with the expectation that when $\bar{C} = C^e$, there will be no net flow into the loop.

Appendix B

In this Appendix, we use the embedding procedure of Brailsford et al. [11] and the SS procedure to calculate the sink strength of a dislocation loop in the effective medium approach. The boundary value problem is defined in eqs. (33) to (37).

Solving this set of equations, we obtain

$$\bar{\psi}(r) = \frac{1}{3}KF(r) + \alpha_1 X(r) - \alpha_2, \quad r_\sigma < r < R, \quad (B.1)$$

$$\bar{\psi}(r) = \frac{K'}{k^2} - \frac{\alpha_3}{r} \exp(-kr), \quad r > R, \quad (B.2)$$

where $F(r)$ and $X(r)$ are defined by eqs. (40) and (41), and $\alpha_1, \alpha_2, \alpha_3$ are integration constants determined by the following equations derived from the boundary conditions

$$-\frac{K'}{k^2} - \frac{\alpha_3}{R} \exp(-kR) = \frac{1}{3}KF(R) + \alpha_1 X(R) - \alpha_2, \quad (B.3)$$

$$\frac{\alpha_3}{R^2} (1 + kR) \exp(-kR) = \frac{1}{g(r)} \left(\frac{KR}{3} - \frac{\alpha_1}{R^2} \right), \quad (B.4)$$

$$\alpha_1 \left(\frac{D}{r_\sigma^2 K_L h(r_\sigma)} + X(r_\sigma) \right) - \alpha_2 + \frac{1}{3}KF(r_\sigma) - \frac{KD r_\sigma}{3\bar{K}_L h(r_\sigma)} + DC^e = 0. \quad (B.5)$$

From eqs. (B.3)–(B.5), we obtain, after eliminating α_2 and α_3 ,

$$\alpha_1 = \left(\frac{1}{3}KF(R) - F(r_\sigma) + \frac{R^2}{g(R)(1+kR)} + \frac{K'}{k^2} + \frac{KDr_\sigma}{3\bar{K}_L h(r_\sigma)} - DC^e \right) \times \left(X(r_\sigma) - X(R) + \frac{D}{r_\sigma^2 \bar{K}_L h(r_\sigma)} + \frac{1}{R(1+kR)g(R)} \right)^{-1}. \quad (\text{B.6})$$

We are now ready to calculate K^e and k^2 . The loss rate, I , of defects into the loop is given by

$$I = -\frac{1}{\Omega} \int_{\sigma} g(r) \bar{\psi}'(r) da + \frac{4}{3}K\pi r_\sigma^3. \quad (\text{B.7})$$

The first term on the right-hand side is the current I_σ into the rate-limiting sphere σ . The second term is the point-defect generation rate inside the rate-limiting sphere. Eq. (B.7) can be obtained by integrating the continuity equation in the volume enclosed between σ and τ . This gives

$$I - I_\sigma = \frac{4}{3}K\pi r_\sigma^3. \quad (\text{B.8})$$

Eqs. (B.1) and (B.7) then gives

$$I = (4\pi/\Omega) \alpha_1. \quad (\text{B.9})$$

For consistency, this must equal the net loss rate per loop in the effective medium which is

$$I = \frac{1}{N_L \Omega} (-K_e + D\bar{C}k^2), \quad (\text{B.10})$$

where $\bar{C}$ is the defect concentration far from the discrete loop. Equating the two currents, we obtain

$$\alpha_1 = K/4\pi N_L \quad (\text{B.11})$$

which gives from eq. (B.6)

$$K_e = k^2 DC^e, \quad (\text{B.12})$$

because from eq. (B.11) α_1 vanishes when K vanishes, and K_e is independent of K . Eliminating α_1 , α_2 , and α_3 from eqs. (B.3)–(B.5) and making use of eq. (B.11), we obtain eq. (39).

It is interesting to note that in the present problem of a single sink type, α_3 can be shown to vanish and hence there is no net current that flows across the boundary separating the effective medium from the

sink free region. This is expected physically because of the equivalence between the effective medium and the discrete sink.

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